Summary

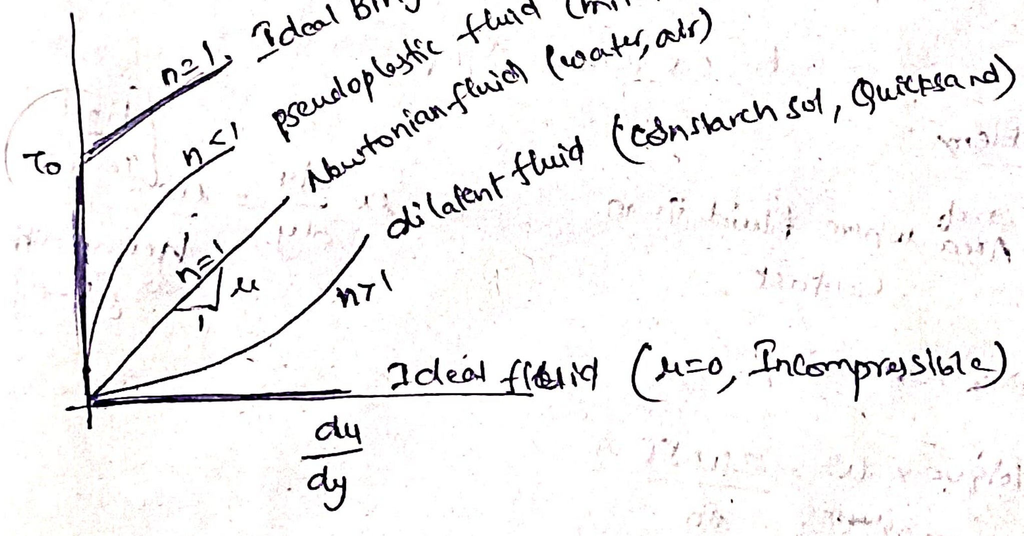
Properties of fluid

1. Density $\rho = \frac{\text{mass}}{\text{Volume}} = \frac{m}{V}$ kg/m³ $\rho_{water} = 1000$ kg/m³

2. Specific weight $\gamma = \frac{\text{Weight}}{\text{volume}} = \frac{W}{V} = \rho g$ * $\boxed{\gamma = \rho g}$
(or)
Weight density

3. Specific gravity
(or)
Relative density $\boxed{S = \frac{\rho_{fluid}}{\rho_{std.fluid}} = \frac{\gamma_{fluid}}{\gamma_{std.fluid}}}$ V. Imp *

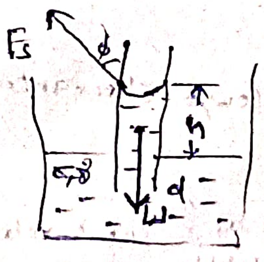
Classification of fluids



Capillarity

$$\boxed{h = \frac{4\sigma\cos\phi}{\gamma d}}$$

surf tension N/m
Angle
sp. weight
dia of capillary tube



Diff in pressure

Drop of liquid = $\boxed{\Delta p = \frac{4\sigma}{D}}$ → surf tension (N/m) → dia of bubble (m)

Liquid Bubble $\boxed{\Delta p = \frac{8\sigma}{D}}$

Cylindrical Jet of liquid $\boxed{\Delta p = \frac{2\sigma}{D}}$

Viscosity

Dynamic/Absolute viscosity (μ): $\underbrace{\tau}_{\text{shear stress}} \propto \underbrace{\frac{du}{dy}}_{\text{velocity gradient}}$

$$\tau = \mu \frac{du}{dy} \Rightarrow \boxed{\mu = \frac{\tau}{\left(\frac{du}{dy}\right)}}$$

$$\boxed{\begin{aligned} 1\ \text{Pa s} &= 10\ \text{poise} \\ 1\ \frac{Ns}{m^2} &= 10\ \text{poise}\end{aligned}}$$

Kinematic Viscosity (ν) $\boxed{\nu = \frac{\mu}{\rho}}$ $\boxed{1\ \text{stroke} = 10^{-4}\ m^2/s}$

Liquids: Temp ↑ Cohesion ↓ Viscosity ↓

Gases: Temp ↑ Molecular Momentum trf ↑ Viscosity ↑

In gases $\mu \propto T^k$ $k = 0.5$ Ideal gas

$k = 0.5 - 0.7$ real gas

V.Imp

*Viscosity problem

1. $\tau = F \times \text{Area}$ (surf Area where fluid is in Contact)

$$\tau = \mu \frac{du}{dy} = \mu \frac{[\text{Final} - \text{Initial Velocity}]}{\text{height diff max } [h - 0]}$$

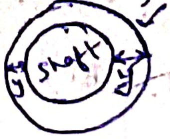
power = Force × velocity

(or)

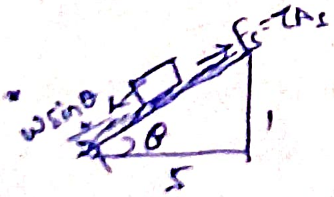
power = Torque × ω = $\frac{2\pi NT}{60}$

Velocity = $\frac{\pi d N}{60}$ (d → dia of shaft, N → speed rpm) (or) $r\omega$

for shaft rotating problems clearance = y



Consider Single y



Terminal Velocity = Const velocity attained by body when state of equilibrium. Always apply Eq^n in prob of Terminal velocity compressibility : $\boxed{\alpha = \frac{-\left\{\frac{d\forall}{\forall}\right\}}{dp} = \frac{\left\{\frac{d\rho}{\rho}\right\}}{dp}}$ $\alpha_{liq} < \alpha_{gas}$

→volume →density →change in pressure

Bulk modulus $\boxed{k = \frac{1}{\alpha}}$

$p \uparrow \quad k \uparrow \quad \alpha \downarrow$

Cavitation $p \leq P_{vap}$

*

Cavitation Number $\boxed{\sigma_c = \frac{p - P_{vap}}{\frac{1}{2}\rho v^2}}$

→Available pr →vap pressure →velocity

$\sigma_c > 0$ Stable cond of liq

$\sigma_c < 0$ Cavitation takes place

~~Rhea~~

Fluid statics

Pressure Intensity of Normal force per unit area $P = \frac{F}{A}$

$1\ pa = N/m^2$ $\quad \boxed{1\ bar = 10^2\ kpa}$ $\quad \boxed{1\ \frac{kgf}{m^2} = 9.81\ \frac{N}{m^2}\ \text{or pa}}$

$kpa = 10^3\ Pa$

$MPa = 10^6\ Pa = N/mm^2$

$GPa = 10^9\ pa$

$$\boxed{\begin{aligned}1\ \text{atm pressure} &= 101.325\ kpa \\ &= 760\ \text{mm of Hg} \\ &= 10.3\ \text{m of water}\end{aligned}}$$

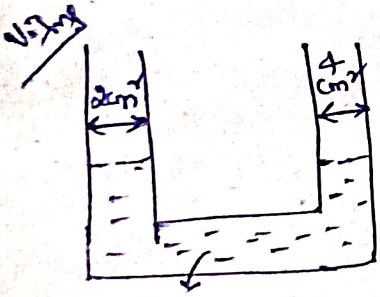
$$\boxed{Head = \frac{Pressure}{Sp.\ Weight} = \frac{P}{\gamma} = \frac{P}{\rho g}}$$

$$\boxed{1\ \text{mm of Hg} \simeq 133\ Pa = 1\ Torr}$$

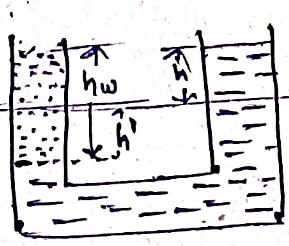
As we go down pressure Increases so + in Manometers
go up press. decreases so −

Manometers

Ex.



water poured in left limb



Initial level.

Vol drop in c/s_1 = Vol fluid rise in c/s_2

$c/s_1\, h' = c/s_2\, h$

$h' = \frac{c/s_2}{c/s_1} \cdot h = \boxed{h' = 2h}$

$V_{water} = c/s_1\, h_w$

$h_w = \frac{30\ cm^3}{2\ cm^2}$

$\boxed{h_w = 15cm}$

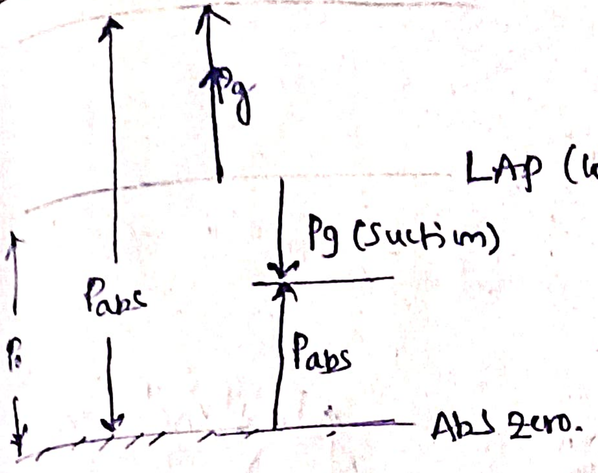
By Isobar

$\gamma_w h_w = \gamma_o (h + h')$

$\gamma_w (15) = 2\gamma_w (h + 2h)$

$6h = 15cm$

$\boxed{h = 2.5cm}$



LAP (local atm pr)

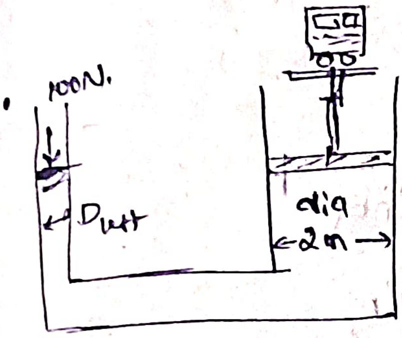
$\boxed{P_{abs} = P_0 + P_g}$

$\boxed{P_{abs} = P_0 - \text{Suction.}}$

P_{abs} = Absolute pr

P_0 = local atm pr (changes from location to location)

P_g = gauge pr



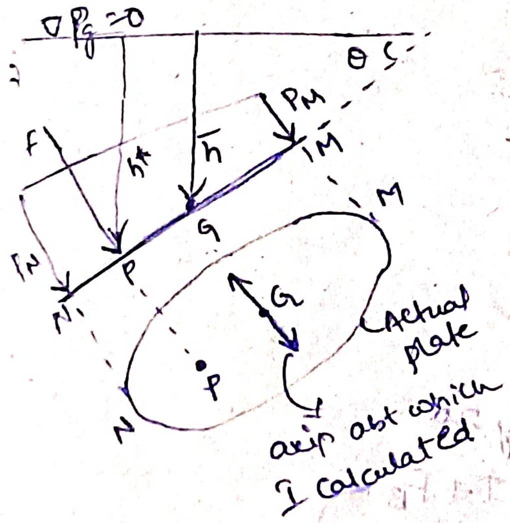
→ pressure applied.

$p A_{2m} = W.$

$$\frac{100}{\frac{\pi}{4} D^2} \cdot \frac{\pi}{4} 2^2 = 10^4$$

$\boxed{D_{left} = 0.2\,m}$

Hydrostatic force



✓ ~~Imp~~

Net hydrostatic force (Total pressure force)

$F = \bar{p} A = p_G A$

$$\boxed{F = \gamma A \bar{h}}$$

- center of pressure (cop) = p

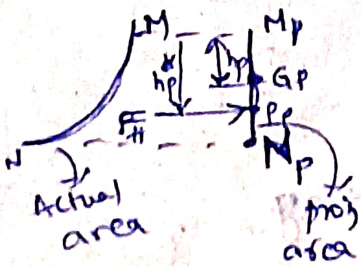
$$\boxed{h^* = \bar{h} + \frac{I \sin^2\theta}{A\bar{h}}}$$

Eccentricity

$$e = \frac{I \sin\theta}{A\bar{h}}$$

Hydrostatic force on curved surf

Hori comp



$$\boxed{F_H = \gamma \{A\bar{h}\}_p}$$

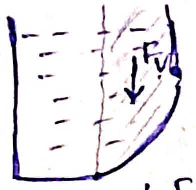
Center of pressure $h_p^* = \left\{\bar{h} + \frac{I}{A\bar{h}}\right\}_p$

Vertical Comp {F_V}

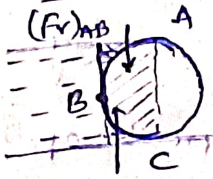
F_V = wt of the fluid

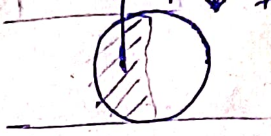
$\boxed{F_V = \gamma \forall}$ → vol of fluid.

Case I Fluid above curved surf

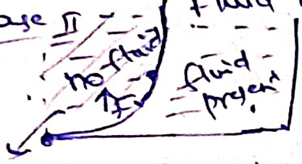


Case III



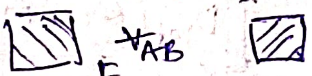


Case II Fluid below curved surf



$F_V = \{(F_V)_{BC} - (F_V)_{AB}\}$ ↑

$F_V = \gamma \forall$ of Imaginary vol



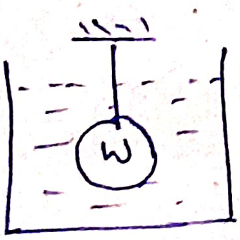
Buoyancy

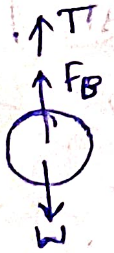
Buoyancy Force F_B = wt of Fluid displaced

$\boxed{F_B = \gamma_f \forall_{\text{fluid displaced}}}$ → Vol of obj present (or) in direct contact in the fluid.

Imp

Apparent weight





By Eqm $\Sigma F_y = 0$

$W = T + F_B$

T = Weight − Buoyance force

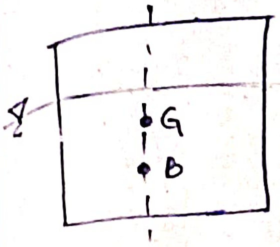
$\boxed{\text{Apparent weight} = W - F_B}$

$\boxed{W = F_B \quad \text{Eq}^m \text{ cond}}$

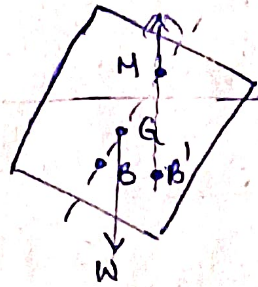
Note: If no wire attached then T = drag

If F_B neglected then ρ_{fluid} can be neglected

Stability of floating Bodies



Initial position



disturbed position

M = metacentre

BM = Metacentric Radius

GM = Metacentric Height

$$\boxed{GM = BM - BG}$$

Waterline

Front view

Top view

pitch

Roll

In this view we need to take I_{min}

BM - metacentric Radius $= \dfrac{I_{min}}{\forall}$ → least I_{min} in top view; $\forall$ ↳ vol displaced (vol contact in fluid)

Metacentric height (GM) $= \dfrac{I}{\forall} - BG$

* Imp

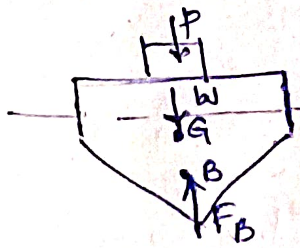
GM > 0	GM < 0	GM = 0
Stable eq^m	unstable Eq^m	Neutral Eq^m

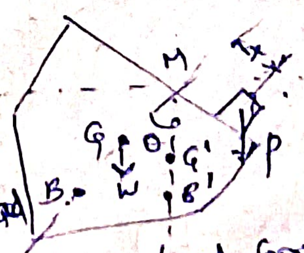
Time period of rolling $T = 2\pi\sqrt{\dfrac{k}{gGM}}$ k = radius of gyration — not of king

$T \propto \dfrac{1}{\sqrt{GM}}$ GM ↑ ⇔ T ↓ Stability ↑ ⇔ comfort ↓

$\{GM\}_{passenger\ ship} < \{GM\}_{warship\ /\ racing\ yatch}$

Experimental method of finding G.M





$$\boxed{(GM..)\ \text{Metacentric height} = \frac{P x}{(W+P)\tan\theta}}$$

P → weight of load; x → displacement of load from ship axis; (W+P) → weight of ship; θ → angle of tilt

Kinematics

Velocity in Cartesian Co-ordinate sym

$$\boxed{\vec{V} = u\hat{i} + v\hat{j} + w\hat{k}}$$ { u, v, w are Comp of velocity in x y z dir }

$\vec{V} = f(x, y, z, t)$

$$u = \frac{dx}{dt} \quad ** $$
$$v = \frac{dy}{dt}$$
$$w = \frac{dz}{dt}$$

$\left.\begin{matrix} u \\ v \\ w \end{matrix}\right\} f(x, y, z, t)$

Ex: $T = 2x^2 + xy + 4t$

v/s $dT = \frac{\partial T}{\partial x}dx + \frac{\partial T}{\partial y}dy + \frac{\partial T}{\partial z}dz$

$$\frac{dT}{dt} = \frac{\partial T}{\partial x}\frac{dx}{dt} + \frac{\partial T}{\partial y}\frac{dy}{dt} + \frac{\partial T}{\partial z}\frac{dz}{dt} + \frac{\partial T}{\partial t}\frac{dt}{dt}$$

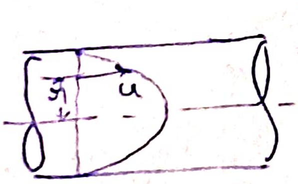
Velocity in Cylindrical polar co-ordinate sym

$$\boxed{\vec{V} = V_r\hat{r} + V_\theta\hat{\theta} + V_z\hat{k}} \rightarrow \vec{V} = f(r, \theta, z, t)$$

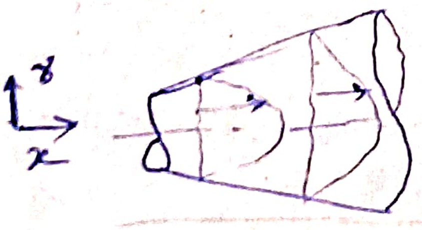
$$V_r = \frac{dr}{dt}, \quad V_\theta = r\frac{d\theta}{dt}, \quad V_z = \frac{dz}{dt}$$

(Radial velocity) (Tangential velocity)

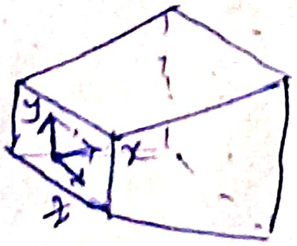
Classification of flow



$u = f(r)$
1D flow
uniform dia pipe



$u = f(r, x)$
2D flow
diverging pipe



Area ↑ along x
By no slip condition u must vary along y & z
$u = f(x, y, z)$
3D flow

$\bar{V} = f(\text{space}, \text{time})$

$t = c$ | Space $= c$

$\bar{V} \neq f(\text{Space})$ — uniform flow → velocity $= c$ w.r.t Space

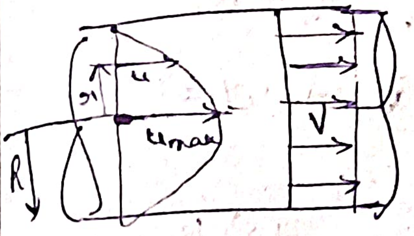
$\bar{V} = f(\text{space})$ — Non uniform flow

$V \neq f(\text{time})$ — Steady flow → V, ρ, P & Temp remains Const w.r.t time

$\bar{V} = f(\text{time})$ — unsteady flow

Discharge (or) Vol flow rate (Q)

mass flux $= \dot{m} = \int d\dot{m} = \int \rho u dA \Rightarrow \boxed{\dot{m} = \rho A V}$



Actual Non unifrom velocity profile

Imaginary uniform V.P

discharge $Q = \int dQ = \boxed{\int u dA}$ $\boxed{Q = A.V}$

$\boxed{Q = \frac{dV}{dt}}$ dV → Volume, dt → time

$\boxed{u = c\{R^2 - r^2\}}$ (c = Const)

u → Velocity at the radius r

R → radius of pipe

r → radius local of velocity

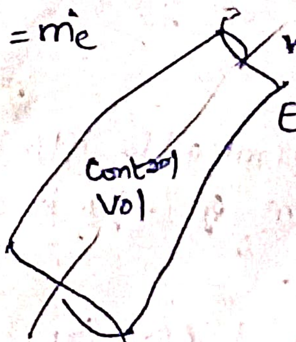
Avg Velocity $\boxed{V = \frac{CR^2}{2} = \frac{U_{max}}{2}}$

Continuity Eqn

Assumptions 1. Steady flow $\rho \neq f(t)$, $\dot{m}_i = \dot{m}_e$

2. flow is In Compressible $\rho \neq f(t)$

3. $A_1 V_1 = A_2 V_2$



$\boxed{\dot{m}_i = \dot{m}_e + \cancel{\frac{\partial m_{cv}}{\partial t}}^{\,0}}$ → rate of storage in Control vol

Differential form of Continuity Eqn in Cartesian Co-ordinate Sym

$$\boxed{\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0.}$$

Rate of change of mass per unit vol in C.V

{Efflux + Influx} an 3 axis / vol in CV.

$$\left\{\frac{\partial m_{cv}}{\partial t}\right\} + \{\dot{m}_e - \dot{m}_i\} = 0$$

$$\boxed{\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0} \Rightarrow \boxed{\overset{\text{divergence}}{\nabla \cdot \bar{V} = 0}} \sim \ Q_i = Q_e$$

$\bar{V} = f(x, y, z)$

If flow is Incomp $\rho = C$, No storage $\frac{\partial \rho}{\partial t} = 0$

Prob $\bar{V} = (5x + 6y + 7z)\hat{i} + (6x + 5y + 9z)\hat{j} + (3x + 2y + \lambda z)\hat{k}$ and $\rho = \rho_0 e^{-2t}$

u , v , w

Sol

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0.$$

$\rho = \rho_0 e^{-2t}$

$\frac{\partial \rho}{\partial t} = -2\rho_0 e^{-2t}$

$$\frac{\partial \rho}{\partial t} + \rho\left\{\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}\right\} = 0$$

$$\boxed{\frac{\partial \rho}{\partial t} = -2\rho}$$

not all at Ism not asking Q's from this topic

Diff form of C.Eqn in Cylindrical polar Co-ordinate Sym

$$\frac{\partial \rho}{\partial t} + \frac{1}{r}(\rho V_r) + \frac{\partial}{\partial r}(\rho V_r) + \frac{1}{r}\frac{\partial}{\partial \theta}(\rho V_\theta) = 0$$

Acceleration

$\bar{a} = \frac{d\bar{V}}{dt}$

$$\boxed{\bar{a} = \frac{\delta V_{time}}{dt} + \frac{\delta V_{space}}{dt}}$$

$\frac{\delta V_{time}}{dt}$ ↓ local Acc or Temp Acc ↓ due to w.r.t time; = 0 in stedy flow

$\frac{\delta V_{space}}{dt}$: Convective Acc ↓ due to w.r.t space; = 0 in uniform flow

Acc in Carterian Co-ordinate Sym

$a_x = \frac{du}{dt}$ $\quad u = f_1(x, y, z, t)$

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz + \frac{\partial u}{\partial t}dt$$

V. Imp

$$a_x = \frac{du}{dt} = \frac{\partial u}{\partial t}\frac{dt}{dt} + \frac{\partial u}{\partial x}\frac{dx}{dt}^{\nearrow u} + \frac{\partial u}{\partial y}\frac{dy}{dt}^{\nearrow v} + \frac{\partial u}{\partial z}\frac{dz}{dt}^{\nearrow w}$$

$$\boxed{\vec{a} = g(x, y, z, t) = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}}$$

$$\boxed{|\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}}$$

Templet

$$\boxed{a \rightarrow \frac{d}{dt} = \frac{\partial}{\partial t} + \left(u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z}\right)}$$

$\frac{\partial}{\partial t}$: Local acc

$\left(u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y} + w\frac{\partial}{\partial z}\right)$: Conv acc

Ex: $a_x = \frac{du}{dt} = \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}$

Flow Visualization

Streamlines

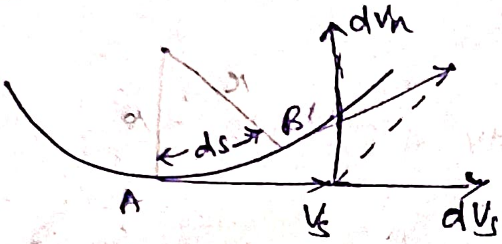
Tangent to sup the direction of velocity $\bar{V}$

$$\boxed{\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}}$$ 3D flow

↓

2D flow

Acceleration w.r.t streamlines :



There are two Comp of acc^n

Tangential Acceleration $a_s = \frac{\partial V_s}{\partial t} + V_s \frac{\partial V_s}{\partial s}$

Normal Acceleration $a_n = \frac{\partial V_n}{\partial t} + \frac{V_s^2}{r}$

Velocity tangential

Normal velocity

Local Acc Convective Acc

$a_s \perp a_n$ $\boxed{\text{Total Acc} = \sqrt{a_s^2 + a_n^2}}$

Pr If flow is steady flow then Local Acc = 0 $V \neq f(t)$)

Pathlines : Curve rep the actual path taken by a particle in a flow.

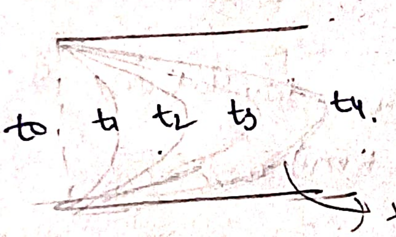
	Particle	time Instant
Streamlines	Various	particular
Pathlines	Particular	Various

Steady flow → pathlines can't Intersect but can overlap

$\boxed{\text{Streamlines} = \text{pathlines} = \text{Streaklines}}$ (for steady flow)

Streaklines

Timelines



Timelines help to Identify Uniform or Non-uniform flow.

Deformation & Rotation

(not Imp)

1. Linear deformation : Rate of linear strain $\dot{\varepsilon}_k = \frac{\partial V_k}{\partial k}$

$$\boxed{\dot{\varepsilon}_x = \frac{\partial u}{\partial x},\ \dot{\varepsilon}_y = \frac{\partial v}{\partial y},\ \dot{\varepsilon}_z = \frac{\partial w}{\partial z}}$$ ★★

Rate of volumetric strain (or) Volumetric dilation rate $= \boxed{\dot{\varepsilon}_x + \dot{\varepsilon}_y + \dot{\varepsilon}_z = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}}$

2. Shear deformation (not Imp)

Avg rate of Shear strain	Total rate of Shear strain	Shear stress
$\dot{\gamma}_{mn} = \frac{1}{2}\left\{\frac{\partial V_n}{\partial m} + \frac{\partial V_m}{\partial n}\right\}$	$\dot{\varepsilon}_{mn} = 2\dot{\gamma}_{mn} = \left\{\frac{\partial V_n}{\partial m} + \frac{\partial V_m}{\partial n}\right\}$	$\tau_{mn} = \mu \dot{\varepsilon}_{mn}$
$\dot{\gamma}_{xy} = \frac{1}{2}\left\{\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right\}$	$\dot{\varepsilon}_{xy} = \left\{\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right\}$	$\tau_{xy} = \mu\left\{\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right\}$
$\dot{\gamma}_{yz} = \frac{1}{2}\left\{\frac{\partial w}{\partial y} + \frac{\partial v}{\partial y}\right\}$	$\dot{\varepsilon}_{yz} = \left\{\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right\}$	$\tau_{yz} = \mu\left\{\frac{\partial w}{\partial y} + \frac{\partial v}{\partial z}\right\}$

If 1D flow with $u(y)$ $\quad \tau_{xy} = \mu\left\{\cancel{\frac{\partial v}{\partial x}} + \frac{\partial u}{\partial y}\right\}$

$$\boxed{\tau_{xy} = \mu \frac{du}{dy}}$$ → Newton's law of Viscosity

V.Imp **

3. Rotation (Q/s won't come from translation & shear come only from rotation)

$$\boxed{\vec{\omega} = \frac{1}{2}\{\bar{\nabla} \times \bar{V}\}}$$

$$\vec{\omega} = f(x, y, z, t) = \omega_x \hat{i} + \omega_y \hat{j} + \omega_z \hat{k}$$

Template for Comp of rotation $\boxed{\omega_k = \frac{1}{2}\left\{\frac{\partial V_{k+2}}{\partial_{(k+1)}} - \frac{\partial V_{k+1}}{\partial_{(k+2)}}\right\}}$

$$\omega_x = \frac{1}{2}\left\{\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\right\}$$

$$\omega_y = \frac{1}{2}\left\{\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x}\right\}$$

$$\omega_z = \frac{1}{2}\left\{\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right\}$$ ✓ V.Imp *

also $\omega_z = 0$ Irrotational flow (2D)

** $\vec{\Omega} = 0$ → Irrotational

max cases $\Omega_z = 0$

4. Vorticity: $\boxed{\text{curl of } \bar{V}}$ is known as Vorticity (Ω, ξ)

* V.Imp *

$$\bar{\Omega} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u & v & w \end{vmatrix}$$

** $\boxed{\vec{\Omega} = 2\vec{\omega} = \{\bar{\nabla} \times \bar{V}\}}$

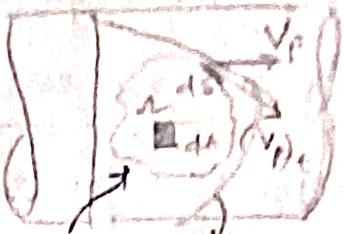
Ω : rad/s

5. circulation $\{\Gamma\}$:

$$\Gamma = \int V_t \cdot ds$$

V_t → Tangential Comp of velocity

ds → length over which V_t is tangential



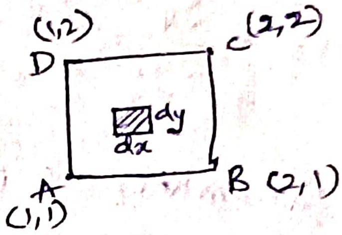
Alternate formula

$$\Gamma = \int \Omega \, dA$$

$$\Gamma = \Omega_{avg} \cdot A$$

$$\Omega_{avg} = \frac{\Gamma}{A}$$

Ex: 2D flow $u = x+y$, $v = x^2 - y^2$



$$\Omega_z = \left\{ \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right\} = 2x - 1$$

$$\Gamma = \int_{y=1}^{y=2} \int_{x=1}^{x=2} 2x - 1 \, dx \, dy$$

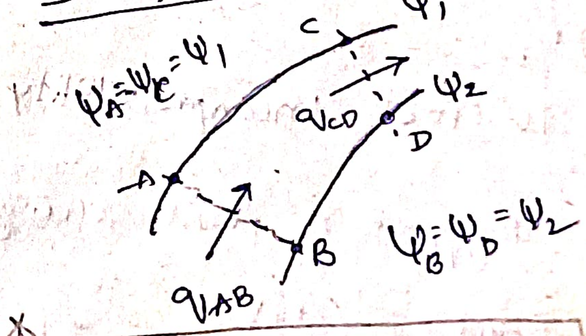
Stream function $\{\psi\}$ (Sye.): It is parent fun of Velocity Vectors, which used to derive U & V in 2D flow. Incompressible

$$u = \frac{-\partial \psi}{\partial y} \quad V = \frac{\partial \psi}{\partial x}$$

(or)

$$u = \frac{\partial \psi}{\partial y} \quad V = \frac{-\partial \psi}{\partial x}$$

- $\psi_A = \psi_C = \psi_1$



$\psi_B = \psi_D = \psi_2$

$$\boxed{\Delta \psi_{\text{accross st line}} = q \text{ (discharge/width)}}$$

$$q_{AB} = |\psi_A - \psi_B| = |\psi_1 - \psi_2|$$

- 2D Incomp flow → Irrotational

Stream fun Satisfy laplace Eq^n

$$\nabla^2 \psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

Velocity potential function $\{\phi\}$ $\longrightarrow$ Irrotational flow

It is a parent fun every comp of velocity can be Identified from the fun

$\phi = f(x, y, z, t)$

$$\boxed{u = -\frac{\partial \phi}{\partial x}, \quad v = -\frac{\partial \phi}{\partial y}, \quad w = -\frac{\partial \phi}{\partial z} \quad \text{(or)} \quad u = \frac{\partial \phi}{\partial x} \quad v = \frac{\partial \phi}{\partial y} \quad w = \frac{\partial \phi}{\partial z}}$$

1. If flow satisfy both (Irrotion + Incompressible) must satisfy Cachy Reiman Eqn.
   (ϕ for Irrotion, ψ for Incompressible)

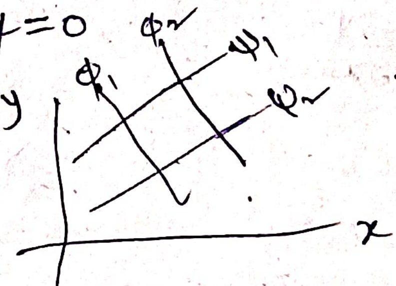
$$\boxed{u = -\frac{\partial \psi}{\partial y} = \frac{\partial \phi}{\partial x} \qquad v = \frac{\partial \psi}{\partial x} = \frac{\partial \phi}{\partial y}}$$

2. Streamlines & Equipotential lines are orthogonal to each other except at stagnation point.

m

- To check Incompressibility the Eqn must satisfy Continuity Eqn

- Stagnation point velocity = 0



$m_1 m_2 = -1$

- $\left(\frac{dy}{dx}\right)_{\text{Stream Lines}} \times \left(\frac{dy}{dx}\right)_{\text{potential lines}} = -1$

- 2D flow $\rightarrow$ translation + rotation + deformation (linear & shear)

- To check Steady flow $\left(\frac{\partial u}{\partial t}, \frac{\partial v}{\partial t} = 0\right)$ — local acc = 0.

- $\nabla \times \vec{V} = 0$ $\{\vec{\Omega} = 0$ Irrotional flow$\}$

  $\nabla \cdot \vec{V} = 0$ $\{$Continuity Eqn In comp flow$\}$

  $\frac{\partial \vec{V}}{\partial t} = 0$ $\{$Steady flow since local acc = 0$\}$

  $\frac{d\vec{V}}{dt} = 0$ $\{$Acceleration = 0$\}$

Fluid Dynamics

Viscous shear force

$$\left\{ \bar{F}_B + \bar{F}_p + \bar{F}_v \right\} = m\bar{a} \quad \{\text{Navier - Stockes Eq}^n\} \rightarrow \text{Conservation of momentum}$$

V.Imp

$$\left\{ \bar{F}_B + \bar{F}_P \right\} = m\bar{a} \quad \{\text{Euler's Eq}^n \text{ of motion}\}$$

Body force   pressure force

Imp: Eulers Eq^n of motion in cartesian co-ordinate sym

$$\boxed{\begin{aligned} \frac{\partial P}{\partial x} &= \rho\{g_x - a_x\} \\ \frac{\partial P}{\partial y} &= \rho\{g_y - a_y\} \\ \frac{\partial P}{\partial z} &= \rho\{g_z - a_z\} \end{aligned}}$$

Template $\frac{\partial P}{\partial k} = \rho\{g_k - a_k\}$

pressure gradient vector

$$\vec{\nabla} P = \left\{ \frac{\partial p}{\partial x}\hat{i} + \frac{\partial p}{\partial y}\hat{j} + \frac{\partial p}{\partial z}\hat{k} \right\}$$

Bernolli Equation

$\rightarrow$ Energy per unit weight form (Head)

$$\boxed{\frac{P}{\rho g} + \frac{V^2}{2g} + z = H}$$

B.Eq. for steady, Incompressible, Inviscid fluids

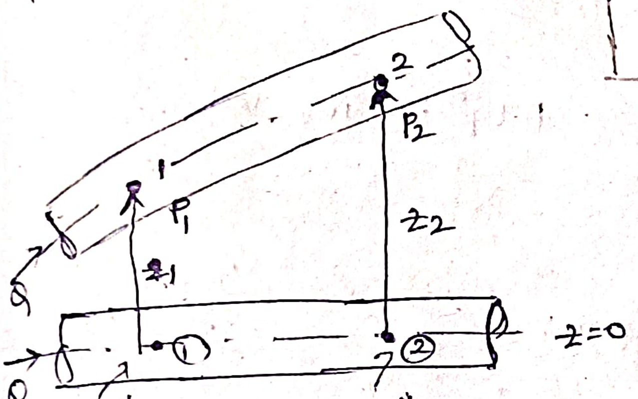
Energy per unit vol form

$$\boxed{\underbrace{P}_{\text{Static pr}} + \underbrace{\frac{\rho V^2}{2}}_{\text{dynamic pr}} + \underbrace{\rho g z}_{\text{Hydrostatic pr}} = \text{Const}}$$

Every term in pr

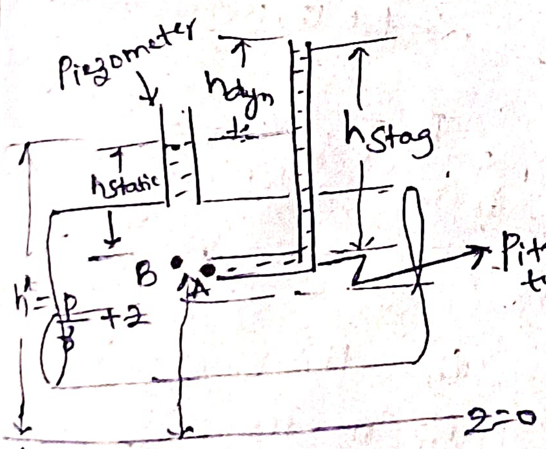
$\left\{P + \frac{1}{2}\rho V^2\right\}$ stagnation pr

$\left\{P^* = \underbrace{P}_{\text{static pr}} + \underbrace{\rho g z}_{\text{Hydrostatic pr}}\right\}$ Piezometric pr



$P_1^* = P_1 + \rho g z_1$ $\quad$ $P_2^* = P_2 + \rho g z_2$

Non-Horizontal flow
$\downarrow$
Effective Horz flow with piezometric pr



$\hookrightarrow$ piezometric head

$$\boxed{h_{stag} = \frac{P}{\gamma} + \frac{V^2}{2g}}$$

$$h_{dyn} = \frac{V^2}{2g}$$

$V_{th} = \sqrt{2gh_{dyn}}$ Theoretical velocity

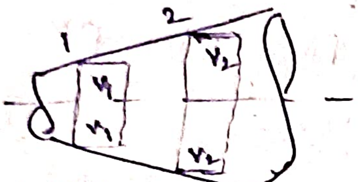
$$\boxed{V_{ac} = \underbrace{C_v}_{\text{Coeff of velocity}} \sqrt{2gh_{dyn}}}$$ Actual velocity

Assumptions in B. Eq^n → Can be applied to rotation & Irrotational

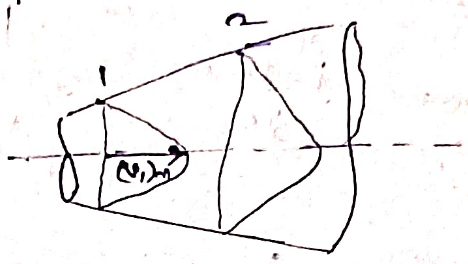
1. It is valid for Ideal flow [originates from Euler's eq^n]
   Steady flow
   Incomp flow [ρ = const]

2. It is valid for along a Stream line

3. BEq^n is applied for section in flow then velocity at individual section must be uniform



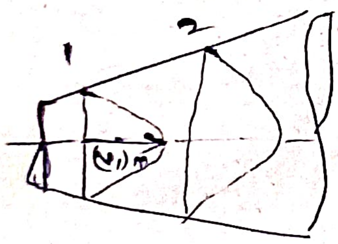
Non uniform flow   B.Eq^n valid



B.Eq^n is Invalid

B.Eq^n in Real flow

1. Velocity profiles are Non-uniform:



Actual velocity Head $= \alpha \frac{V^2}{2g}$

$$\boxed{\alpha = \frac{1}{AV^3}\int u^3 \cdot dA}$$

α = KE Correction factor

$$\frac{P_1}{\gamma} + \alpha_1 \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\gamma} + \alpha_2 \frac{V_2^2}{2g} + z_2$$

2. Involvement of Hydraulic m/c

a. Turbine

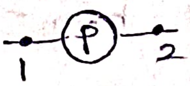
$H_1 > H_2$

$$\boxed{H_1 = H_2 + H_T}$$

$$\frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + z_2 + H_T$$

b. pump

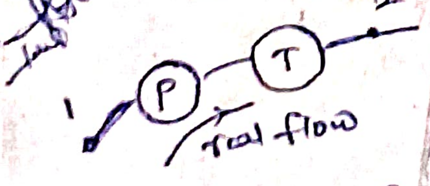
$H_1 < H_2$



$$\boxed{H_1 = H_2 - H_P}$$

$$\frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1 = \frac{P_2}{\gamma} + \frac{V_2^2}{2g} + z_2 - H_P$$

3. Involvement of Viscous effect. ** In real flow



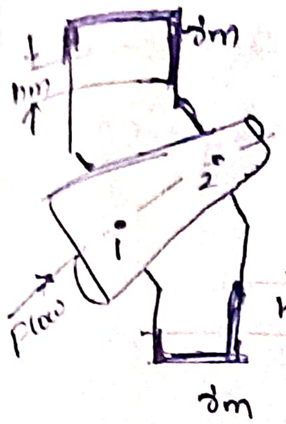
$$\underbrace{\frac{P_1}{\gamma} + \alpha_1 \frac{V_1^2}{2g} + z_1}_{H_1} = \underbrace{\frac{P_2}{\gamma} + \alpha_2 \frac{V_2^2}{2g} + z_2}_{H_2} - H_P + H_T + h$$

H_P: pump head; H_T: Turbine Head; h: head loss (m. of water)

$H_1 > H_2$

Y V Singh

Conversion of diff manometric head into diff head of W.f:



Inverted U Tube manometer

$\gamma_m < \gamma_w$ $\boxed{h_w = h_m \left\{ 1 - \frac{S_m}{S_w} \right\}}$

$\gamma_m << \gamma_m$ $h_w \simeq h_m$

U-Tube Manometer

$\gamma_m > \gamma_w$

$\boxed{h_w = h_m \left\{ \frac{S_m}{S_w} - 1 \right\}}$

$\gamma >> \gamma_w$

$h_w \simeq h_m \left\{ \frac{S_m}{S_w} \right\}$

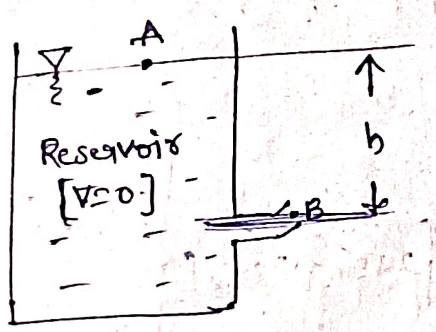
Imp Eq.n of power $\boxed{P = \rho Q g H = \gamma Q H}$

$P_{loss} = \gamma Q h_{loss}$

$P_{Turbine} = \gamma Q H_T$

$P_{pump} = \gamma Q H_p$

Impo
Force liq Jet



By B.Eq.n b/w A & B

$$\frac{P_A}{\gamma} + \frac{V_A^2}{2g} + Z_A = \frac{P_B}{\gamma} + \frac{V_B^2}{2g} + Z_B$$

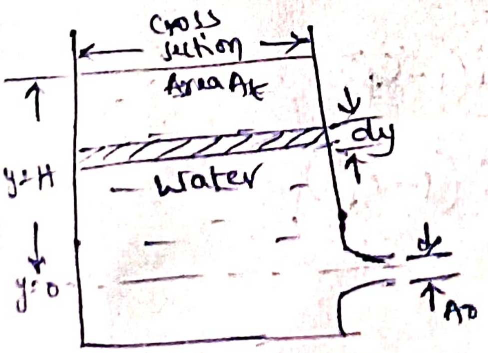
$V_B = \sqrt{2g(Z_A - Z_B)}$

V.Imp $\therefore \boxed{V_B = \sqrt{2gh}}$

Time Taken to empty tank

As time progress value y changes but V=C at orifice

Approach: time taken for small drop [dy] for case V=C.



$dV = A_t(-dy)$

Velocity at t $= \sqrt{2gy}$

$Q = A_0\sqrt{2gy}$

$\boxed{Q = \frac{dV}{dt} \Rightarrow dt = \frac{dV}{Q}}$

$\int y^{-\frac{1}{2}} \Rightarrow \frac{y^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = \frac{y^{1/2}}{\frac{1}{2}}$

$\int y^{-\frac{1}{2}} \Rightarrow 2\sqrt{y}$

$dt = \frac{-A_t\, dy}{A_0\sqrt{2gy}}$

$\int_0^t dt = \frac{-A_t}{A_0\sqrt{2g}} \int_H^{H/4} y^{-1/2}\, dy$

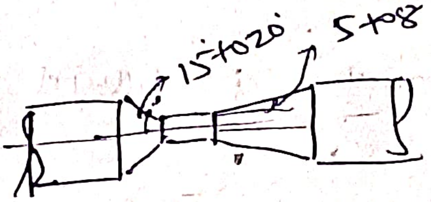
Velocity Measurement

Effect of pitot tube + piezometer → Pitot Static tube (or) Prandtl's pitot tube

(stagnation head) (static head)

Discharge (or) flow measurements

1. orifice meter ($C_d = 0.60 - 0.65$)
2. venturi meter ($C_d = 0.95 - 0.98$)
3. Nozzle meter ($C_d = 0.8 - 0.85$)



1. Theoretical discharge $Q_{th} = \{\text{Geometry term}\} \times \{\text{diff pr term}\}$

Geom	× Δp^* term
$\dfrac{A_1 A_2}{\sqrt{A_1^2 - A_2^2}}$	$\sqrt{2g\Delta h^*}$
	$\sqrt{2g h_m \left\{\frac{S_m}{S_w} - 1\right\}}$ (or) $\sqrt{2g h_m \left\{1 - \frac{S_m}{S_w}\right\}}$
$\dfrac{\pi}{4} \dfrac{D_1^2 D_2^2}{\sqrt{D_1^4 - D_2^4}}$	$\sqrt{\dfrac{2\Delta p^*}{\rho}}$

→ Pr diff b/w entry & throat (Pa)

2. Head loss in obstruction flow meter

~~$h = \Delta h^*(1 - C_d^2)$~~

$$\boxed{h = \Delta h^* (1 - C_d^2)}$$

$$\boxed{Q_{ac} = C_d\, Q_{th}}$$

↳ discharge coeff

Vortex motion

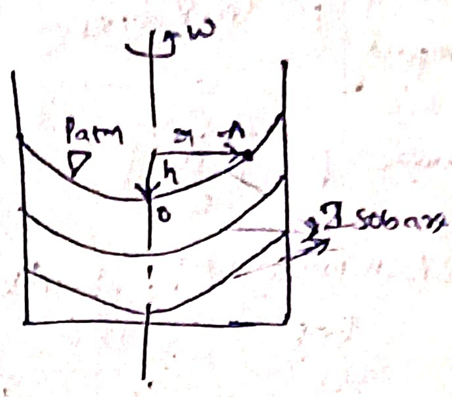
Old results in forced vortex motion

$$\frac{P_0}{\gamma} - \frac{V_0^2}{2g} + z_0 = \frac{P_A}{\gamma} - \frac{V_A^2}{2g} + z_A$$

$$z_A = \frac{V_A^2}{2g}$$

$$\boxed{h = \frac{r^2\omega^2}{2g}}$$

not Imp I think
not came in
previous GATE paper



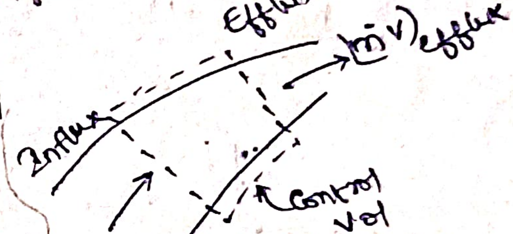
V. Imp
Momentum eqn and Applications (2m sure)

$$\vec{F} = \boxed{\left\{(\dot{m}\vec{v})_{efflux} - (\dot{m}\vec{v})_{Influx}\right\}} + \boxed{\frac{\partial(\dot{m}\vec{v})_{cv}}{\partial t}}$$

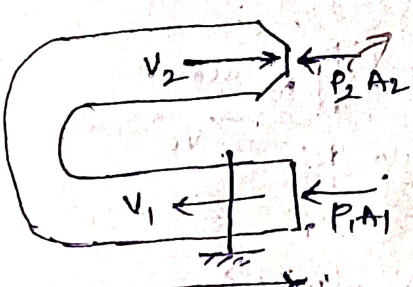
$\frac{\partial(\dot{m}\vec{v})_{cv}}{\partial t} \to 0$ for steady flow

$\dot{m}\vec{v}$ = momentum flux (N)

Rate of storage of momentum in Control vol



Ex



$P_2 = 0$ open to atm

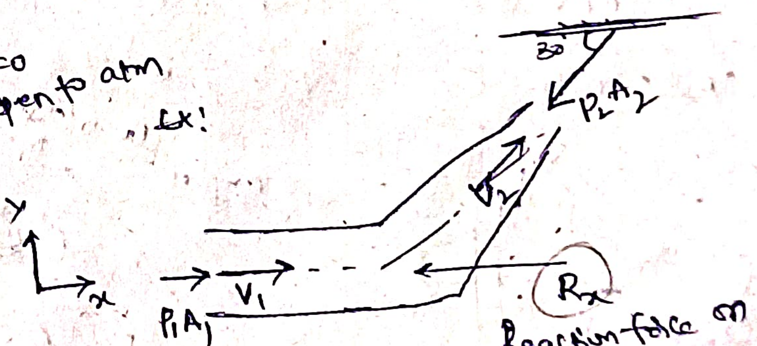
Reaction force → F_x

$$\Sigma F_x = (\dot{m}v_x)_2 - (\dot{m}v_x)_1$$

$$-P_1A_1 - \cancel{P_2A_2} + F_x = \dot{m}\left[V_2 - (-V_1)\right]$$

Exposed to atm pr
$P_2 = 0$

Ex!



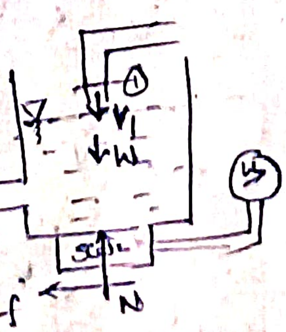
Reaction force on bend pipe

$$\Sigma F_x = (\dot{m}v_x)_2 - (\dot{m}v_x)_1$$

$$P_1A_1 - R_x - \cancel{P_2A_2}\cos 30° = \dot{m}\left\{V_2\cos 30 - V_1\right\}$$

Exposed to atm
$pr = 0$

Ex!



$\dot{m} = \rho Q$

$$\boxed{\begin{aligned}\Sigma F_y &= \cancel{(\dot{m}v_y)_2} - (\dot{m}v_y)_1 \\ N - W &= -\rho Q(-V_1)\end{aligned}}$$

$$\boxed{\begin{aligned}\Sigma F_x &= (\dot{m}v_x)_2 - \cancel{(\dot{m}v_x)_1} \\ -\mu N &= \rho Q(-V_2)\end{aligned}}$$

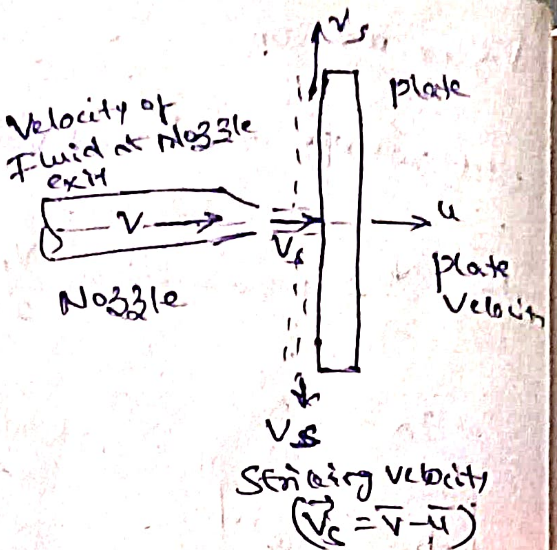
Impact of Jets

Force of Impact $\boxed{\vec{F} = (\dot{m}\vec{V})_{striking} - (\dot{m}\vec{V})_{exit}}$

power $\boxed{P = \vec{F}\,\vec{u}}$

$\dfrac{\text{K.E at Nozzle exit}}{\text{time}} = \boxed{\frac{1}{2}\dot{m}V^2 = \frac{1}{2}\rho A V^3}$

Efficiency of Jet $\boxed{\eta = \dfrac{\vec{F}\vec{u}}{\frac{1}{2}\rho A V^3}}$



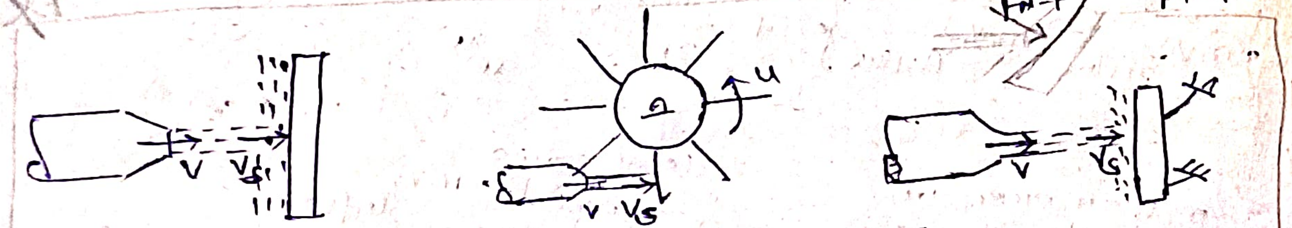
(Friction on surf Neglected)

a) $V_{exit} = \bar{V} - \bar{u}$

b) Net force = Normal to plate

$F_N = F$

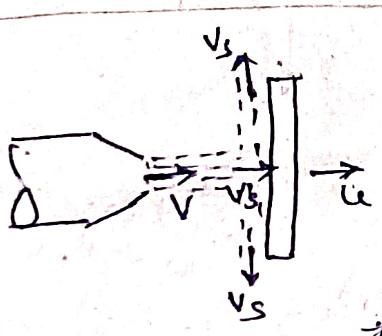
Striking Mass flux



$\dot{m}_{st} = \rho A(\bar{V} - \bar{u})$ | $\dot{m}_{st} = \rho A V$ | $\dot{m}_{st} = \rho A V$

$V_{st} = \bar{V} - \bar{u}$ | $V_{st} = \bar{V} - \bar{u}$ | $V_{sf} = V$

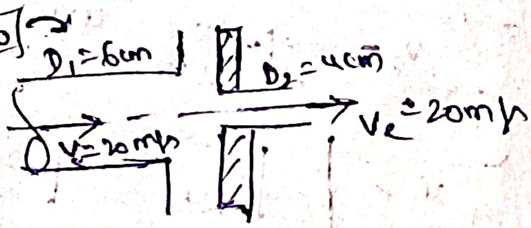
1)



$F_x = (\dot{m}V_x)_{st} - \cancel{(\dot{m}V_x)_{exit}}$

$= \rho A(V-u)(V-u)$

ex prob $\boxed{u=0}$

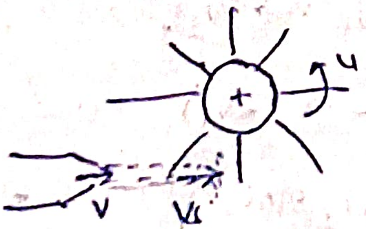


$F_x = (\dot{m}V_x)_{st} - (\dot{m}V_x)_{exit}$

$F_x = V\{\dot{m}_{st} - \dot{m}_{exit}\}$

$= V\{\rho A_{st}V - \rho A_{exit}V\}$

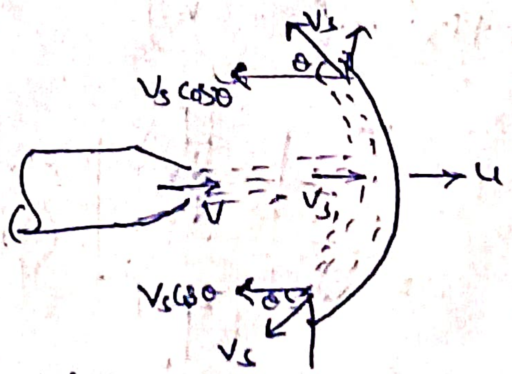
2)



$F_x = (\dot{m}V_x)_{st} - (\dot{m}V_x)_{exit}$

$F_x = \rho A V(V-u)$

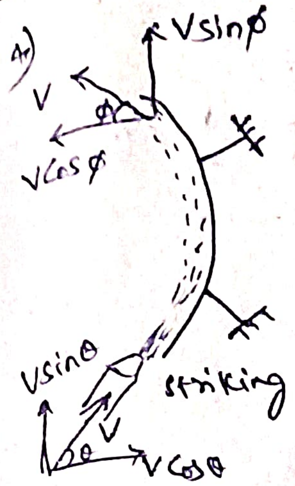
3.



$F_x = (\dot{m}V_x)_{SF} - (\dot{m}V_x)_{exit}$

$F_x = \rho A(V-u)\{(V-u) - -(V-u)\cos\theta\}$

4)



$F_x = (\dot{m} V_x)_{st} - (\dot{m} V_x)_{exit}$

$F_x = \dot{m}(V_{x\,st} - V_{x\,exit})$

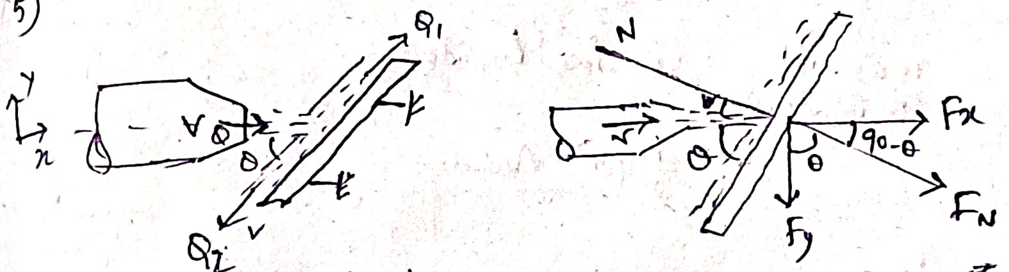
$\boxed{F_x = \rho A V [V\cos\theta - (-V\cos\phi)]}$

$F_y = (\dot{m} V_y)_{st} - (\dot{m} V_y)_{exit}$

$= \dot{m}\{V_{y\,st} - V_{y\,exit}\}$

$\boxed{F_y = \rho A V \{V\sin\theta - V\sin\phi\}}$

5)



Eqll of Force in tangential dir

$F_T = (\dot{m} V_T)_{st} - (\dot{m} V_T)_{exit} = 0$

$(\dot{m} V_T)_{st} = (\dot{m} V_T)_{exit}$

$\rho Q V\cos\theta = \rho Q_1 V - \rho Q_2 V$

$\boxed{\begin{aligned} Q_1 - Q_2 &= Q\cos\theta \\ Q_1 + Q_2 &= Q \end{aligned}}$ → solve for

$F_N = (\dot{m} V_N)_{st} - \cancel{(\dot{m} V_N)_{exit}}$

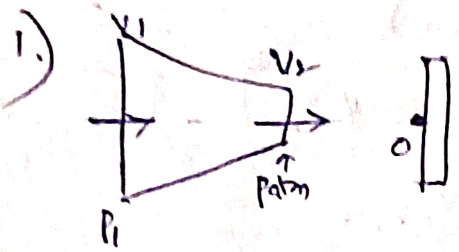
$\boxed{F_N = \rho A V . V\sin\theta}$

$\boxed{\begin{aligned} F_x &= F_N \cos(90-\theta) \\ F_y &= F_N \cos\theta \end{aligned}}$

$\boxed{\begin{aligned} Q_1 &= \frac{Q}{2}\{1+\cos\theta\} \\ Q_2 &= \frac{Q}{2}\{1-\cos\theta\} \end{aligned}}$

problem

1.)

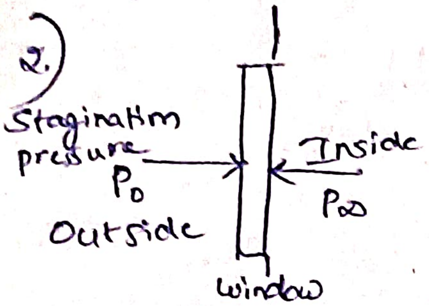


Pressure at point o : $P_o = \frac{1}{2}\rho V_2^2$

or

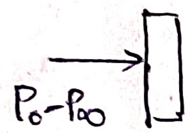
Dynamic pressure

2.)



$P_0 = P_\infty + \frac{1}{2}\rho V^2$

$P_0 - P_\infty = \frac{1}{2}\rho V^2$



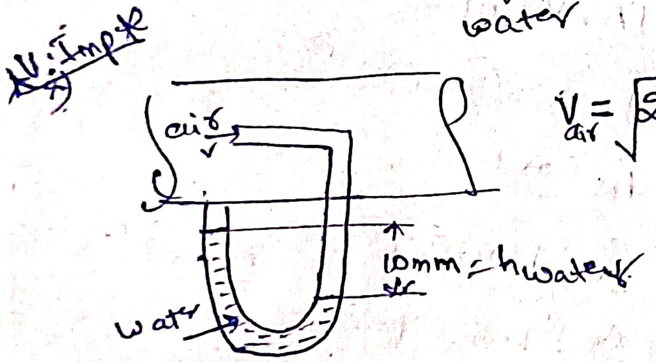
Force act on window $= (P_0 - P_\infty) A_{window}$

$= \frac{1}{2}\rho V^2 \times A_{window}$

3) relative density $= \frac{\rho_{liq}}{\rho_{ref\ sub}}$

water

4) Imp



$V_{air} = \sqrt{2 g h_{air}}$

$h_{air} = h_{water}\left[\frac{S_{water}}{S_{air}} - 1\right]$

manometric fluid

Internal flow

Reynold's Num = $\dfrac{\text{Inertia force}}{\text{Viscous force}}$

$$\boxed{Re = \frac{\rho V L}{\mu} = \frac{V \cdot L}{\nu}}$$ → Ch. length

1. Flow through pipe

$$\boxed{Re = \frac{\rho V D}{\mu} = \frac{V D}{\nu}}$$

$Re \leq 2000$ Laminar

$2000 < Re < 4000$ Transitional

$Re \geq 4000$ Turbulent

2. Flow through duct (Non-circular c/s)

$$\boxed{D_h = \frac{4A}{P}}$$

A → c/s area of flow

P → wetted perimeter

D_h → Ch. length

Standard results in every Internal flow through pipe

1. $r\left[\dfrac{-dp}{dx}\right] = \dfrac{d(r\tau)}{dr}$

2. $\left\{\dfrac{-dp}{dx}\right\} = \dfrac{\Delta P}{L}$ ✓

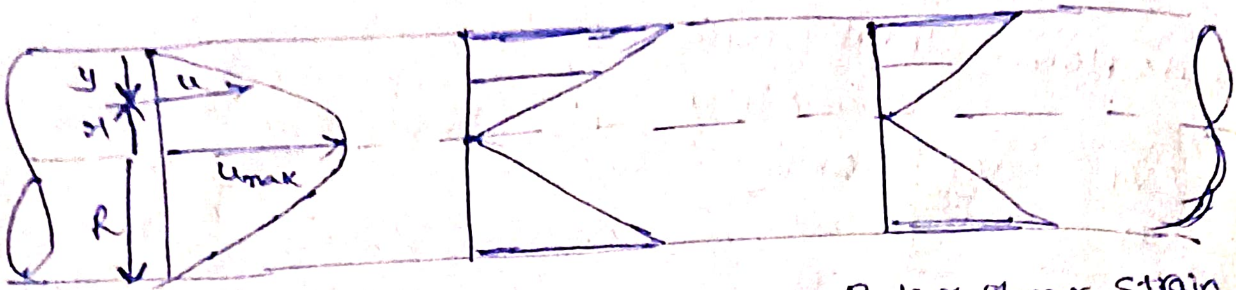
** 3. $\tau = \left(\dfrac{-dp}{dx}\right)\dfrac{r}{2}$ ✓ shear stress

4. $f' = \dfrac{\tau_0}{\frac{1}{2}\rho V^2}$, $f = 4f'$

5. $\boxed{h = \dfrac{\Delta p}{\gamma} = \dfrac{f L V^2}{2 g D}} = \dfrac{1}{12.1}\dfrac{f L Q^2}{D^5}$ → Darcy weisbach eq^n ✓

h → major head loss

Laminar flow through pipe



Velocity profile

• Parabolic

$\alpha = 2$

$\beta = \frac{4}{3} = 1.33$

Shear stress

$$\boxed{\tau = \left(\frac{-dp}{dx}\right)\frac{r}{2}}$$

Linear variation

Rate of shear strain

$$\tau = \mu \frac{du}{dy}$$

$$\frac{du}{dy} = \frac{1}{\mu}\left[\frac{-dp}{dx}\right]\frac{r}{2}$$

Linear variation

Formula

1. Local velocity $\boxed{u = \frac{1}{4\mu}\left\{\frac{-dp}{dx}\right\}(R^2 - r^2)}$ Imp

2. Max. local velocity $\boxed{U_{max} = u|_{r=0}}$ $U_{max} = \frac{1}{4\mu}\left(\frac{-dp}{dx}\right)R^2$

3. Avg velocity of flow $\boxed{V = \frac{U_{max}}{2}}$ $V = \frac{1}{8\mu}\left[\frac{-dp}{dx}\right]R^2$

from this eq

4. Discharge $Q = AV$

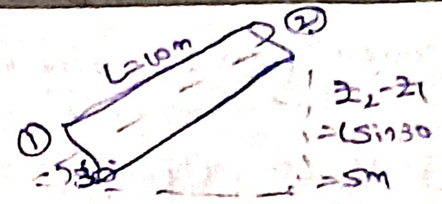
5. Pressure drop $\boxed{\frac{-dp}{dx} = \frac{\Delta P}{L}}$ $\boxed{\Delta P = \frac{32\mu V L}{D^2}}$ Avg vel $V = \frac{1}{8\mu}\frac{\Delta P}{L}R^2$

6. Head loss $\boxed{h = \frac{\Delta P}{\gamma} = \frac{f L V^2}{2gD}}$

7. Local skin friction Coeff $\boxed{f' = \frac{\tau_0}{\frac{1}{2}\rho V^2} = \frac{16}{Re}}$

8. Darcy friction factor $\boxed{f = \frac{64}{Re}}$

ex: $\Delta p \neq \frac{32\mu VL}{D^2}$ since pipe is non horizontal



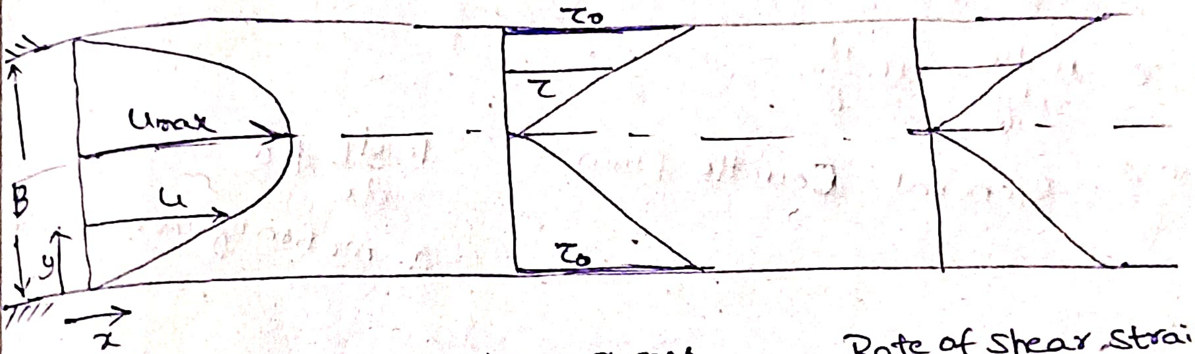
But $\Delta p^* = \frac{32\mu VL}{D^2}$

$$\Delta p^* = p_1^* - p_2^*$$

$$\Delta p^* = (p_1 + \rho g z_1) - (p_2 + \rho g z_2)$$

$$\Delta p = p_1 - p_2 = \Delta p^* + \rho g (z_2 - z_1)$$

Laminar flow b/w 2 Fixed ||el Flat plates:



Velocity profile
- Parabolic

$\alpha = 1.54$

$\beta = 1.20$

Shear stress
- Linear profile

$\tau = \mu \frac{du}{dy}$

Rate of shear strain
- Linear profile

$\frac{du}{dy} = \frac{\tau}{\mu}$

Formulas

1. Local velocity $\boxed{u = \frac{1}{2\mu}\left\{\frac{-dp}{dx}\right\}(By - y^2)}$

2. Max Local velocity $\boxed{U_{max} = u\Big|_{y=B/2}}$ $\quad U_{max} = \frac{1}{8\mu}\left\{\frac{-dp}{dx}\right\}B^2$

3. Avg velocity of flow $\boxed{V = \frac{2}{3}U_{max}}$ $\quad V = \frac{1}{12\mu}\left(\frac{-dp}{dx}\right)B^2$

4. Discharge/unit width $\boxed{q = VB}$ $\quad q = \frac{1}{12\mu}\left(\frac{-dp}{dx}\right)B^3$

5. pressure drop $\boxed{\left\{\frac{-dp}{dx}\right\} = \frac{\Delta p}{L}}$ $\quad V = \frac{1}{12\mu}\frac{\Delta p}{L}B^2 \Rightarrow \boxed{\Delta p = \frac{12\mu VL}{B^2}}$

6. Head loss $\boxed{h = \frac{\Delta p}{\gamma}}$ $\quad h = \frac{12\mu VL}{\gamma B^2} = \frac{12\mu qL}{\gamma B^3}$

7. Shear stress $\boxed{\tau = \mu\frac{du}{dy}}$

Couette flow : Laminar flow b/w $\parallel^{le}$ plat plates → one moving, 1 Fixed

Case 1 : simple/pure/plain couette flow $\left(\frac{dp}{dx}=0\right)$

velocity at y dist from bottom

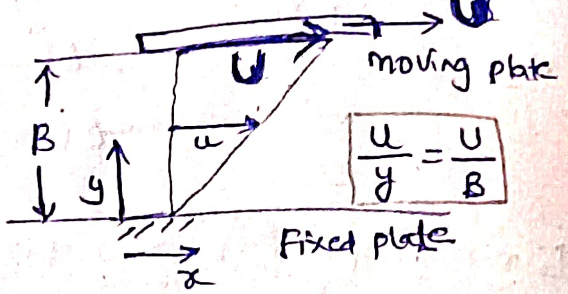
1. $\boxed{u=\frac{U}{B}\cdot y}$

Avg velocity

2. $V=\frac{U}{2}$

3. $q=\frac{U}{2}\times B$

4. $\tau=\mu\frac{du}{dy}=\mu\cdot\frac{U}{B}$



Case 2 : General Couette flow

1. $\frac{dp}{dx}\neq 0$
2. motion of moving plate

Formulas : General C.F = Simple C.F + Laminar flow b/w plates

Flow through pipes

Total head loss h = Major Head loss + Minor head loss
in a flow

$$\boxed{h = h_f + \Sigma h_{min}}$$

Major head loss (h_f).

Case I: Laminar flow $f = \frac{64}{Re}$

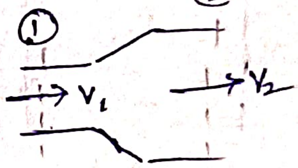
$$\boxed{h_f = f\frac{LV^2}{2gD}} = \boxed{\frac{1}{12.1} f \frac{LQ^2}{D^5}}$$

**

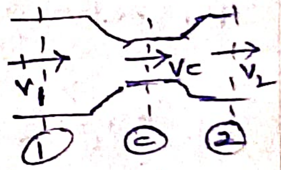
$$\boxed{h_f \propto \Delta P}$$

Minor Head loss (h_{min})

1. Sudden expansion $\boxed{h_{exp} = \frac{(V_1 - V_2)^2}{2g}}$



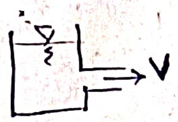
2. Sudden Contraction $\boxed{h_{cont} = \frac{V_2^2}{2g}\left\{\frac{1}{C_c} - 1\right\}^2}$



$$\boxed{C_c = \frac{V_2}{V_c}}$$

3. Exit loss $\boxed{h = \frac{V^2}{2g}}$

4. Entry loss $\boxed{h = \frac{1}{2}\frac{V^2}{2g}}$



5. If coeff of head loss (K) $\boxed{h_{min} = K\frac{V^2}{2g}}$

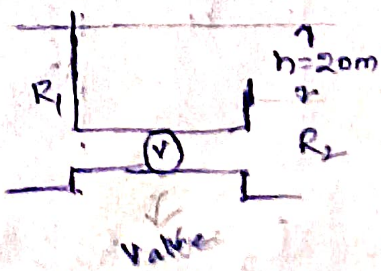
Ex: In Bends, Fittings, Entry, exit

Q: $h = h_f + \Sigma h_{min}$

$$h = \frac{fLV^2}{2gD} + \left\{h_{entry} + h_{valve} + h_{exit}\right\}$$

$$20\,m = \frac{fLV^2}{2gD} + \left\{0.5\frac{V^2}{2g} + 5.5\frac{V^2}{2g} + \frac{V^2}{2g}\right\}$$

Coeff head loss given in Q



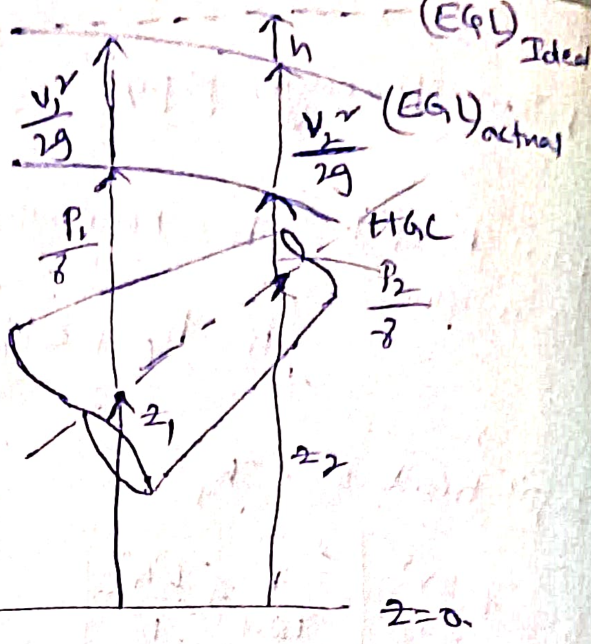
Gradient Lines

$(EGL)_{Ideal} - (EGL)_{act}$: Head loss

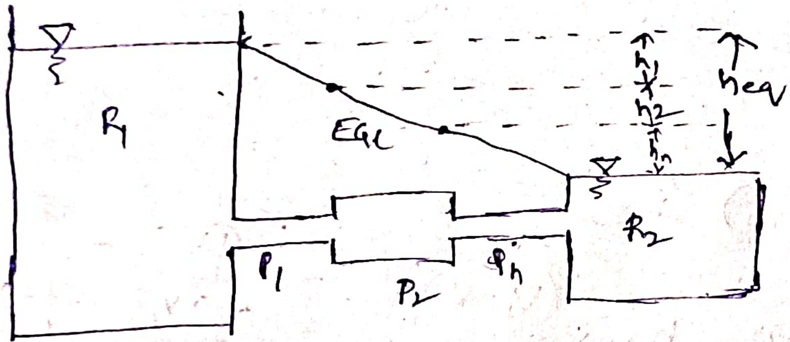
$(EGL)_{act} - (HGL)$ = Dynamic Head (or) Velocity Head

EGL rises up due to pump

EGL rises down due to turbines losses



Series Connection:



By continuity Eq" $\boxed{Q_1 = Q_2 = Q_3 \dots Q_n = Q_{eq}}$

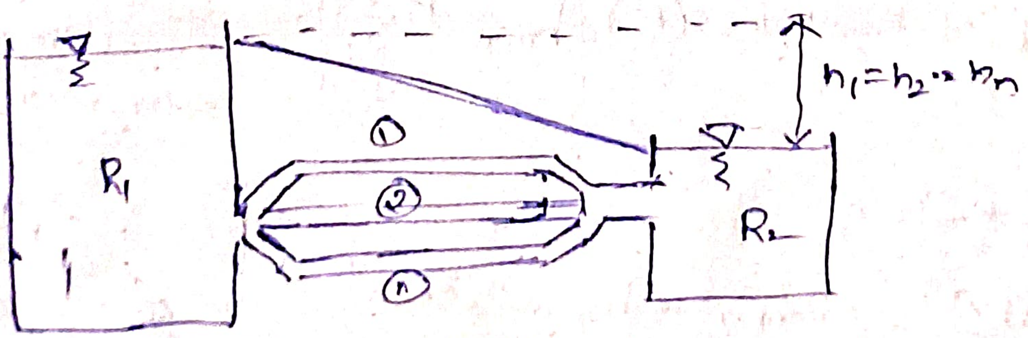
By energy eq^n: $\boxed{h_{eq} = \sum_{i=1}^{n} h_i = h_1 + h_2 + \dots h_n}$

$$\boxed{\frac{L}{D^5} = \sum_{i=1}^{n}\left(\frac{L}{D^5}\right)_i = \left(\frac{L}{D^5}\right)_1 + \left(\frac{L}{D^5}\right)_2 \dots \left(\frac{L}{D^5}\right)_n}$$

f=c

Dupits eq^n

Parallel Connection:



$$\boxed{Q_{eq} = \sum_{i=1}^{n} Q_i = Q_1 + Q_2 \dots Q_n}$$

$$\boxed{h_{eq} = h_1 = h_2 \dots = h_n}$$